

GAME DESIGN DOCUMENT

EcoRangers

“Saving the planet, one piece of trash at a time.”

1. Game Overview

1.1. Game Summary

Title: EcoRangers
Genre: 2D Puzzle, Arcade, Educational
Platform: PC (Windows), Web (Mobile Support)
Target Audience: Children age 9-12
Development Team: 5 person
Development Time: 4 months
Engine: Unity

EcoRangers is a fast-paced 2D arcade, educational puzzle game where players must keep up with an endless stream of waste and sort each item correctly. Different types of trash require different treatments before they can be properly sorted. The goal is to manage the waste efficiently, earn the highest score possible before time runs out.

1.2. Design Motivation

Most trash-sorting games focus only on sorting waste, overlooking what happens before it reaches that stage. This makes waste processing feel simpler than it actually is.

EcoRangers expands the idea of trash sorting by turning waste preprocessing into part of the gameplay. Players learn that different types of waste require different preparation before they can be sorted correctly. The goal is to make learning about waste management feel natural and fun without making it feel like a lesson.

1.3. Design Pillars

The game is designed around five primary pillars:

1. Sorting Accuracy

The player should learn to identify and correctly categorize different types of waste rather than simply dragging items into the nearest bin. Each item requires the player to recognize its waste category and determine the appropriate disposal method.

2. Waste Processing

The game should introduce the idea that some waste cannot be sorted immediately. Certain items may require additional actions, such as cleaning, separating components, or processing them before they can be placed in the

correct bin. This adds depth to the sorting mechanic while teaching proper waste-handling practices.

3. Time Management

The player must balance speed and accuracy while working against a continuously decreasing timer. Correctly sorting waste provides additional time, encouraging players to maintain a steady pace without sacrificing accuracy.

4. Progressive Learning

The game should gradually introduce more complex waste items and processing requirements. Early gameplay focuses on recognizing basic waste categories, while later stages challenge players to apply more detailed knowledge and handle items that require additional steps.

5. Engagement and Replayability

The combination of scoring, combos, multipliers, increasing difficulty, and a high-score target encourages players to replay the game and improve their performance. The fast-paced arcade structure keeps the learning experience interactive and enjoyable rather than presenting waste management as a purely instructional activity.

2. Gameplay

2.1. Gameplay Flow

In EcoRangers, players are tasked with sorting various types of trash that appear on a conveyor belt. Inspired by Flash-era games and casual mobile games, EcoRangers features simple and familiar gameplay: players drag each piece of trash from the conveyor belt into the correct bin.

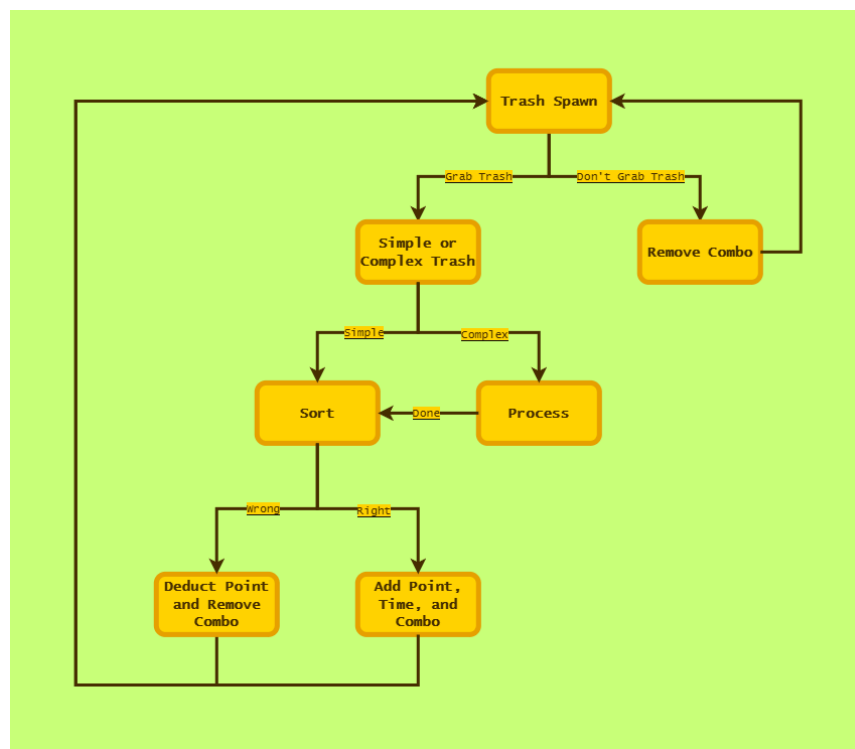
As the game progresses, the trash becomes more complex. Some items may require additional steps or processing before they can be properly sorted. This adds depth to the otherwise simple trash-sorting gameplay while also educating players about the proper ways to handle and process different types of waste.

Correctly sorting an item awards the player points and adds a small amount of time to the timer. Consecutive correct sorts build a combo, increasing the score multiplier every 3 correct sorts, up to a maximum of 5x. Sorting an item incorrectly deducts points and resets the multiplier. Allowing a trash item to pass without sorting it only resets the multiplier.

A countdown timer continuously decreases throughout the game. The player must reach their high score target before the timer runs out.

2.2. Core Gameplay Loop

The primary gameplay loop consists of:



3. Mechanics

EcoRangers features several mechanics:

3.1. Basic Sorting

The player must sort each piece of trash into the correct category. Trash will appear on a conveyor belt, and the player must quickly identify and move each item to its appropriate category.

3.2. Trash and Bins

The main components of the game are trash and bins. Trash appears on the conveyor belt on the left side of the screen, while the bins are placed on the right. The player must drag each piece of trash into the correct bin before it disappears.

Correctly sorted trash awards points and additional time. Incorrectly sorted trash deducts points and removes the score multiplier. Missing a piece of trash also resets the player's score multiplier.

3.3. Tray

A tray is placed in the middle of the screen and can hold one piece of trash at a time. It can be used as temporary storage as well as an essential component for more complex trash-processing mechanics.

3.4. Complex Trash

3.4.1. B3 Type Trash

Inspired by Indonesian waste processing and categorization, B3 waste refers to hazardous waste that requires special handling. Before being disposed of in the correct bin, B3 waste must first be placed inside a plastic bag.

When a piece of B3 waste is dragged onto the plastic bag, it is converted into bagged waste and placed on the tray. This means the tray must be empty before the B3 waste can be processed.

3.4.2. Residual Type Trash

Inspired by Indonesian waste processing and categorization, residue waste refers to inorganic waste that is difficult to recycle. Before being disposed of, residue waste must first be wrapped in paper.

When residue waste appears, the player must drag the paper onto the trash. The waste will then be converted into wrapped waste and placed on the tray. This means the tray must be empty before the residue waste can be processed.

3.5. Composite Trash

3.4.1. Water Bottles

Water bottles may appear with their cap and label still attached. The player must first place the bottle on the tray to begin processing.

Once the bottle is on the tray, the player can separate the cap, label, and bottle and place each component into its respective bin.

3.4.1. Sealed Box

Sealed boxes may appear and must first be placed on the tray. The player must then remove the tape by dragging it off the box. The tape must be disposed of in a separate bin before the box can be processed further.

Once the tape has been removed, the player must flatten the box by pressing it three times. After the box has been flattened, it can be disposed of in the appropriate bin.

3.4.1. Sauce Bottles

Sauce bottles may appear with their cap, label, and sauce residue still intact. The player must first place the bottle on the tray before processing it.

Once on the tray, the player must separate the cap and label and dispose of each in its respective bin. The player must then drag the bottle, which still contains sauce residue, into the sauce bucket to empty it.

After the sauce has been removed, the bottle becomes a clean bottle and is placed back on the tray. The clean bottle can then be disposed of in the appropriate bin.

4. Score and Progression

4.1. Score System

Each correctly sorted piece of trash awards the player points, with more complex trash processing providing higher scores. An incorrect sort deducts points and resets the score multiplier. The player's objective is to achieve the highest possible score before the timer runs out.

Trash Type	Trash Components		Score	Additional Time
Basic	Red Apple		10	1.5 seconds
	Green Apple		10	1.5 seconds
	Chicken Thighs		10	1.5 seconds
	Leaf		10	1.5 seconds
	Blue Bottle		10	1.5 seconds
	White Bottle		10	1.5 seconds
	Paper		10	1.5 seconds
	Cardboard		10	1.5 seconds
Complex	B3	Battery	15	1.5 seconds
		Medicine	15	1.5 seconds
		Light Bulb	15	1.5 seconds
	Residue	Gum	15	1.5 seconds
		Cigarette	15	1.5 seconds
Composite	Water Bottle	Bottle	20	1.5 seconds
		Cap	15	0.5 seconds
		Label	15	0.5 seconds
	Sealed Box	Tape	15	1seconds
		Cardboard	30	1.5 seconds
	Sauce Bottle	Bottle	25	1.5 seconds
		Cap	15	0.5 seconds
		Label	15	0.5 seconds

4.2. Combo and Multiplier

Successfully sorting three consecutive pieces of trash increases the player's score multiplier, up to a maximum of 5x. Incorrectly sorting trash or missing a piece of trash resets both the combo and the multiplier.

4.3. Difficulty Progression

Players begin with basic sorting, where trash is directly sorted into the correct bin. As they reach score milestones, they progressively unlock more complex types of trash that require the tray for additional processing and sorting.

Score Milestone	Unlocked Trash Type	Description
0	Basic Trash	Trash is directly sorted into the correct bin
200	Bottle and Paper	Inorganic trash is seperated into bottles and papers
500	B3 Trash	B3 waste must first be placed inside a plastic bag before bing disposed of
	Residue Trash	Residue waste must first be wrapped in paper before being disposed of
1000	Water Bottle (5% chance)	The cap, label, and bottle must be seperated before the each component is disposed
2000	Sealed Box (5% chance)	The tape be seperated, then the box must be flattened before being disposed of
3000	Sauce Bottle (5% chance)	The cap, label, and bottle must be seperated, then the bottle must be emptied first before being disposed of
5000	-	Increase all Composite Trash chance from 5% to 10 %
10000	-	Increase all Composite Trash chance from 10% to 20 %

5. Education and Learning Design

5.1. Background

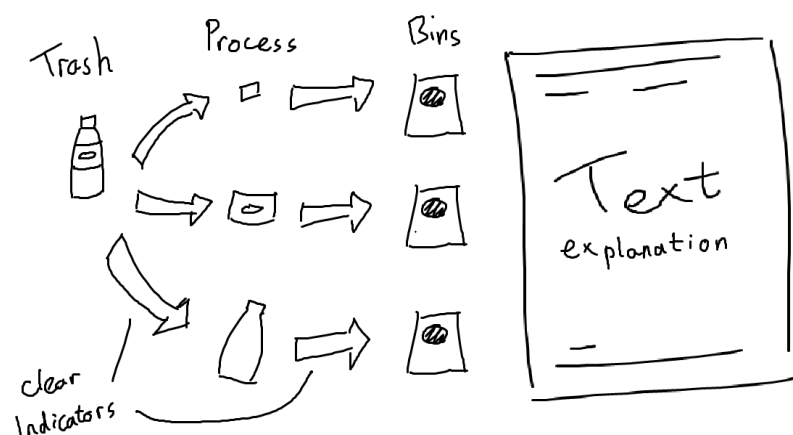
Waste management remains a major environmental challenge in Indonesia, driven by population growth, increasing consumption, and limited public awareness of proper waste sorting. The large amount of waste generated each year is not fully managed, leading to environmental and health risks. This situation highlights the importance of introducing waste management and sorting habits from an early age. Therefore, EcoRangers was developed as a technology-based educational game that provides children with a simple, interactive, and enjoyable way to learn about proper waste sorting and environmental responsibility.

EcoRangers aims to develop an interactive game-based educational medium to improve children's understanding and awareness of the importance of properly sorting and managing waste from an early age. Specifically, the program seeks to increase children's knowledge of different types of waste through educational content, develop their ability to sort waste according to its type through interactive games, foster positive habits related to waste sorting and management, provide an interactive, enjoyable, and easy-to-understand learning platform, and integrate waste management education with digital game-based technology.

5.2. Tutorial Design

The tutorial should prioritize teaching players about different types of trash and the proper way to process and sort them. It should use clear visuals and concise explanations to demonstrate each step, helping players understand what the trash is, how it should be processed, and which bin it belongs in.

Visual indicators should guide players through the process step by step, while short text provides essential educational context about the material and its correct disposal. The goal is for players to learn proper waste-sorting practices through gameplay, rather than simply memorizing the actions required to complete each task.



6. UI / UX

6.1. UI/UX Direction

The UI uses a familiar, clean, and cartoonish art style inspired by casual and mobile games. Its visual language is designed to feel cohesive with the overall game, creating a fun and playful experience while keeping navigation simple and intuitive.

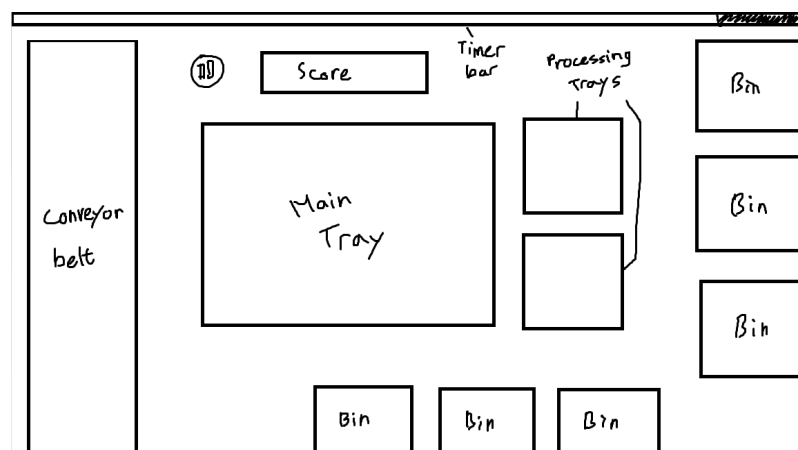
6.2. Main Menu

The main menu features a lush green environment with scattered trash, creating a calming yet engaging atmosphere. A bright blue sky and a distant city skyline provide visual contrast against the natural backdrop, helping the title stand out while reinforcing the game's environmental theme.

6.1. Gameplay

During gameplay, all key UI elements are positioned at the top of the screen to keep the play area clear and focused. The score UI pulses whenever the score changes, providing immediate visual feedback. Combo counters and multipliers further enhance the sense of progression and reward successful actions.

The timer is represented as a horizontal bar that gradually decreases as time passes. When the remaining time is low, the bar pulses red to clearly communicate urgency and encourage quick decision-making.



Tutorial cards appear between segments to brief players on the upcoming task. Each card provides clear and concise instructions for the required trash-sorting and processing actions. Subtle pulses and visual highlights draw attention to important elements, helping players quickly understand new mechanics.

7. Art and Visual

7.1. Art Style Overview

The game uses 2D stylized casual/mobile game art with a clean, colorful, and cartoonish appearance. The visual direction is designed to feel cheerful, approachable, and playful, with an emphasis on simplicity and immediate readability.

The overall visual identity is characterized by:

- Bright, saturated colors
- Simple geometric forms with soft and rounded shapes
- Simple gradients and highlights with minimal details
- Friendly and clean atmosphere

The artwork should feel friendly, playful, and immediately readable, as the game is targeted toward children. Objects and characters should be recognizable at a glance and remain visually distinct even when presented at small sizes.

7.2. Art Direction

The art direction aims to achieve the following:

7.2.1. Readability

As the game is targeted towards children, the most important goal is clarity. The players should quickly understand which objects can be interacted with.

7.2.2. Friendly and Approachable

The game is intended to feel light and welcoming, with the use of rounded and smooth shapes, bright and saturated colors, and simple highlights.

7.2.3. Consistency

Objects, environments, UI elements, and effects should share the same clean and cartoonish visual language.

7.3. Color Direction

The primary palette consists of high-saturation colors for the objects and UI elements, while the environments use light and low-saturation colors.

The color palettes are:



7.4. Trash Design

The trash should use a clean, cartoonish art style inspired by casual and mobile game visuals. The design should prioritize clarity, readability, and immediate recognition.

Trash should generally have:

- Rounded and soft shapes
- Limited detail and simple shading
- Strong, easily recognizable silhouettes

As the core visual content of the game, trash objects should remain visually prominent and easy to identify. Each object should have a distinct silhouette and recognizable identity, allowing players to quickly understand and sort it.

7.5. Trash Bins Design

Trash bins serve as the sorting destinations for all trash types and should be immediately distinguishable from one another. Each bin should use a unique category color and include a small thumbnail representing the type of trash it accepts. The visual relationship between each trash type and its corresponding bin should be clear and intuitive, allowing players to identify the correct destination at a glance without unnecessary thought or confusion.

7.6. Environment Art

The environment should provide context and establish a clean setting without competing with interactable objects.

The environment should:

- Use light and low-saturated colors
- Maintain low visual contrast
- Use simple texture and minimal decoration

Overall, the environment should support the gameplay experience while keeping interactable objects as the primary visual focus.

7.7. Shading and Lighting

The game primarily uses a combination of simple cel-shading and soft-shading to create a clean, stylized visual appearance. Shadows should use clear, simplified cel-shaded transitions, while highlights remain subtle to add dimensionality without creating excessive contrast. Gradients should be used to communicate changes in value and form. Overall, the shading should remain simple and controlled, balancing defined shadows, subtle highlights, and soft value transitions to give objects a sense of volume while maintaining the clean, cartoon-like aesthetic.

7.8. Text and Typography

The typography should remain clear and highly readable while maintaining the game's playful visual style. Text uses a bold, thick treatment with distinct outlines to ensure strong visibility against different backgrounds

Important information can use subtle pulsing animations to attract the player's attention without becoming distracting. For high scores, milestones, and new progress, text can also pulse with changes in color, providing immediate visual feedback and reinforcing a sense of achievement and reward.

8. Music and Audio

8.1. Audio Direction

The audio design aims to create a cheerful, calm, and friendly atmosphere that complements the game's educational and approachable nature. Audio should provide clear feedback for player actions while maintaining a consistent overall tone. Each sound is designed to make interactions feel responsive and intuitive without becoming distracting.

8.2. Music

The game features two main music tracks, both sourced from JDSherbert's free music asset pack on itch.io.

The menu music uses cheerful and friendly instrumentation, with guitar and drums creating a welcoming atmosphere that complements the game's bright visual environment. Its purpose is to create a positive first impression and encourage players to begin playing.

The gameplay music is calmer and more subdued. The slower guitar-based track creates a cozy and relaxing atmosphere, allowing players to remain focused while sorting and processing trash.

8.3. Sound Effects

All sound effects are sourced from [pixabay](https://pixabay.com), and are used to provide immediate feedback for player actions and make interactions feel more responsive and immersive. Each major gameplay interaction has its own distinct sound effect, allowing players to understand what is happening through both visual and auditory feedback.

Gameplay sound effects include:

- Grabbing trash.
- Placing trash on the tray.
- Placing B3 trash into a plastic bag.
- Wrapping residue trash with paper.
- Placing composite trash on the tray for processing.
- Unboxing and flattening cardboard boxes.

- Emptying sauce bottles.
- Placing trash into the correct bin.
- Correct and incorrect sorting, with distinct sounds to clearly communicate the result.
- Achieving a high score.
- A ticking sound that plays when time is almost up.
- A continuous conveyor belt sound to reinforce the game's processing environment.

UI interactions also have supporting sound effects, such as panels swooping down and buttons being clicked. These sounds provide additional feedback and help make the interface feel responsive and cohesive with the gameplay experience.

9. Playtesting and Iteration

We conducted several playtesting sessions for EcoRangers to evaluate the gameplay experience and identify areas for improvement. Based on feedback and observations from these sessions, we made several design changes to improve the overall player experience.

Problem	Observation	Design Change	Result
Trash cycle in the game is too slow	In the early stages, players can quickly sort through the trash, resulting in excessive waiting time.	When there is no trash on screen, immediately spawn a new piece of trash regardless of the cooldown.	When players sort trash quickly, they can immediately sort additional trash, rewarding faster players.
	The trash moves too slowly along the conveyor, making it easy to grab and causing the gameplay to feel stalled.	Increase the trash movement speed to make gameplay faster and more engaging.	Trash moves along the conveyor faster, making the early stages more engaging while keeping the later stages challenging.
Timer display is not intuitive	Displaying the remaining time as text takes longer for players to process at a glance.	Replace the timer text with a visual timer bar so players can easily see how much time is remaining.	Players can easily visualize the remaining time at a glance, allowing them to focus on sorting.
Score is positioned in the middle of the screen	There is already a lot of activity in the center of the screen, making the score distracting and difficult to monitor.	Move the score to the top of the screen alongside the pause button and timer UI.	Players can check the score and multipliers without becoming distracted during sorting. They can also see the remaining time, making the gameplay more intuitive.
Trash bins take up too much space	There are a total of nine trash bins, making it difficult for all of them to fit neatly along the right side of the screen.	Move several trash bins to the bottom of the screen and distribute them between the right and bottom areas.	The trash bins are clearer and more spaced out, making it easier for players to drag trash into them.
Timer feels too short for some players	Processing complex and composite trash can take longer to complete, while the additional time awarded is too limited.	Increase the overall timer from 60 seconds to 90 seconds.	The gameplay lasts longer, allowing players to reach later stages and achieve higher scores.
		Increase the amount of additional time awarded from composite trash.	Processing composite trash is worth it because sorting it correctly grants more additional time, making the extra processing time beneficial rather than wasteful.
Tutorial is unclear for some players	The tutorial has limited space, resulting in a very simple diagram that does not fully explain the more complex steps.	Add small and short explanatory text to the diagram to clarify what players need to do during the more complex steps.	Players can understand the diagram more easily.

10. Development Process

10.1. Team

Our team is composed of 5 people, each with their own role and tasks.

Member	Role
Adriel Kevin Jonathan	Game Designer, Programmer
Darryl Arief Tananjaya	Programmer, QA
James Tiono	Writer, Researcher
Jocelyn Tania Harjanto	2D Artist, QA
Kristoforus Vinan Adriska	Writer, Researcher

10.2. Schedule

EcoRangers was developed over approximately four months, from March to June 2026. To plan and manage the development timeline, we used a Gantt chart to organize tasks, set milestones, and track the overall schedule.

For day-to-day development and progress tracking, we adopted a Kanban-based SDLC workflow. This allowed the team to clearly monitor the status of tasks, coordinate responsibilities, and identify work that needed attention. Jira was used as the primary project management tool to manage tasks and track each team member's progress throughout development.

